Modelling and Forecasting the US Consumer Price Index: A Seasonal ARIMA Approach

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Abstract. This report analyses the US Consumer Price Index for All Urban Consumers (CPI-U), a monthly price index published by the Bureau of Labor Statistics, using observations from January 2010 to December 2019. The first 108 observations (2010–2018) are used for model building; the final 12 (2019) are withheld for forecast evaluation. Following a Box-Cox variance stabilisation step ($\hat{\lambda} = 1.229$), deterministic detrending is shown to be inadequate, and regular first differencing ($d = 1$) combined with seasonal differencing ($D = 1$, period 12) is used to achieve stationarity. ACF/PACF analysis leads to five candidate SARIMA models. Model selection via AIC and BIC identifies $\text{ARIMA}(0,1,1)(0,1,1)_{12}$ — the classical airline model — as the best specification, outperforming the competing $\text{ARIMA}(1,1,0)(0,1,1)_{12}$ suggested by `auto.arima` on all information criteria, all residual diagnostics, and every out-of-sample accuracy metric, achieving a MAPE of 0.37% on the 2019 test set.

1. Introduction

Prices affect every corner of economic life. Whether a household is planning its weekly grocery budget, an employer is negotiating wages, or a central bank is deciding whether to raise interest rates, tracking how the overall cost of living changes over time is essential. The Consumer Price Index for All Urban Consumers (CPI-U), published monthly by the US Bureau of Labor Statistics (BLS), is the most widely used measure of consumer-price inflation in the United States. It tracks the average change in prices paid by urban consumers for a fixed basket of goods and services — food, energy, housing, medical care, and more — and is indexed so that the average price level during 1982–84 equals 100.

The CPI-U matters well beyond academic interest. It is used to adjust Social Security payments, to index tax brackets, to set the coupon on inflation-protected government bonds (TIPS), and as a benchmark in countless wage contracts. Being able to forecast it even a few months ahead therefore has real practical value for policymakers and households alike.

2. Model Specification

2.1. Data and Exploratory Analysis

The full CPI-U dataset spans January 1913 to the present (1,332 monthly observations, was obtained from:[1].). This study uses the sub-sample January 2010 – December 2019 (120 observations). The last 12 months (2019) form the test set; the preceding 108 months (2010–2018) form the training set. Summary statistics for the training series are given in Table 1.

Table 1: Summary statistics — CPI-U training series (Jan 2010 – Dec 2018).

Min	Q1	Median	Mean	Q3	Max
216.69	229.30	236.10	235.10	241.40	252.89

Figure 1 shows the training series. There is a clear, near-linear upward trend throughout the period — reflecting steady but modest inflation in the US economy — with variance appearing roughly constant and a mild seasonal oscillation visible at the annual frequency.

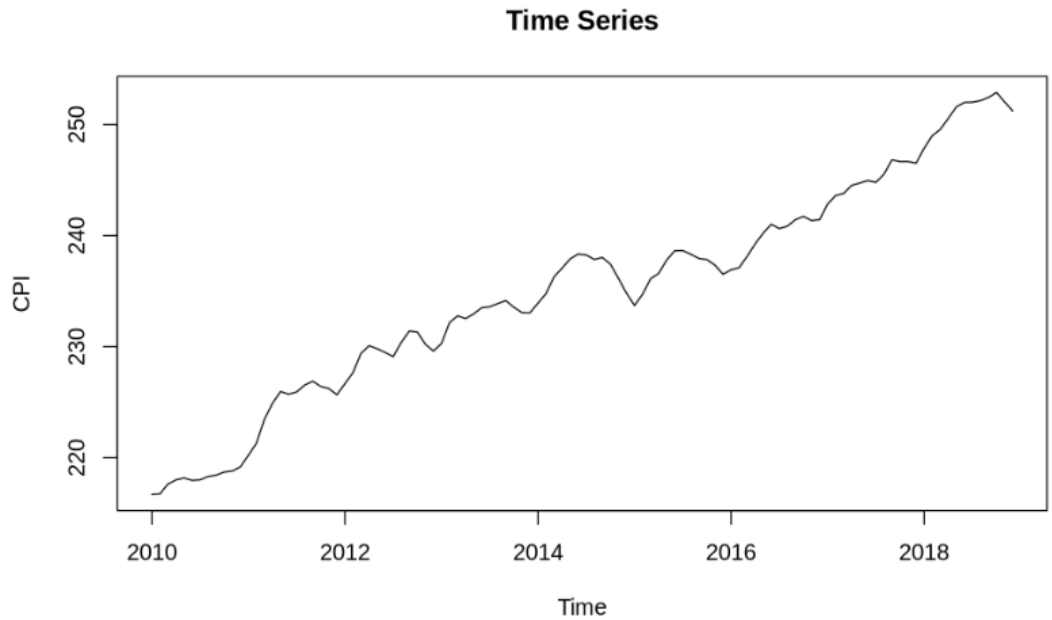

Figure 1: Monthly CPI-U training series (Jan 2010 – Dec 2018). The series exhibits a clear upward trend consistent with low, steady inflation over the decade.

2.2. Variance Stabilisation

Although the variance appears visually stable, a Box-Cox transformation was applied as a formal precaution. The Guerrero method estimates $\hat{\lambda} = 1.229$, close to 1, confirming that no strong transformation is required. The transformation is nonetheless retained throughout to guard against subtle heteroscedasticity.

2.3. Stationarity Testing

Level series. The Augmented Dickey-Fuller (ADF) test on the Box-Cox-transformed series yields $DF = -2.70$ ($p = 0.288$): the null of a unit root is not rejected, confirming non-stationarity in line with the visible trend.

Deterministic detrending. Two deterministic trend models were explored. A linear regression of the transformed series on time ($R^2 = 0.958$) and an extension that adds an annual sinusoidal harmonic ($R^2 = 0.963$) were both estimated. In both cases the residuals failed all three white-noise tests simultaneously: the ADF test still failed to reject the unit-root null; the Shapiro-Wilk test rejected normality ($p < 0.001$); and the Runs test strongly rejected independence ($p < 10^{-16}$). Deterministic detrending was therefore judged insufficient.

First differencing. Applying one regular difference to the Box-Cox series gives $DF = -4.67$ ($p = 0.01$), rejecting the unit root and establishing $d = 1$.

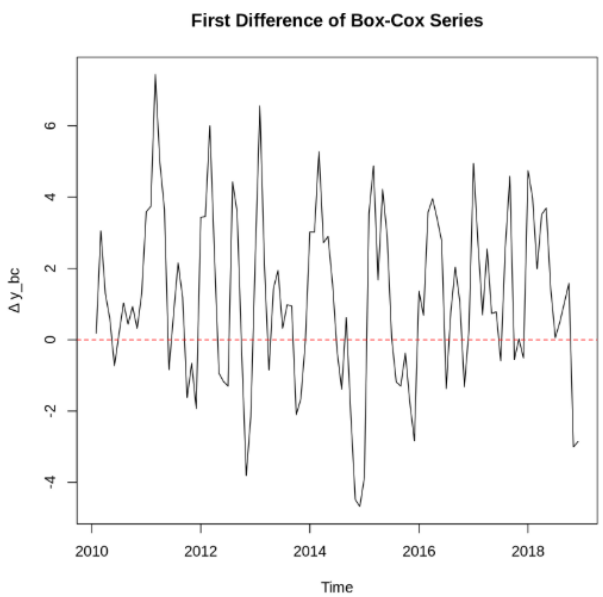

Figure 2: First difference of the Box-Cox CPI-U series (Δy_{bc}). The trend is removed; the series fluctuates around zero, but seasonal patterns remain.

2.4. ACF/PACF Analysis and Seasonal Differencing

The ACF and PACF of the first-differenced series (Figure 3) show prominent spikes at lags 12, 24, 36, …, indicating residual seasonality with period 12. A seasonal difference (lag 12) is applied on top of the regular difference; the ADF test on the doubly differenced series confirms stationarity ($DF = -3.73$, $p = 0.026$), establishing $D = 1$.

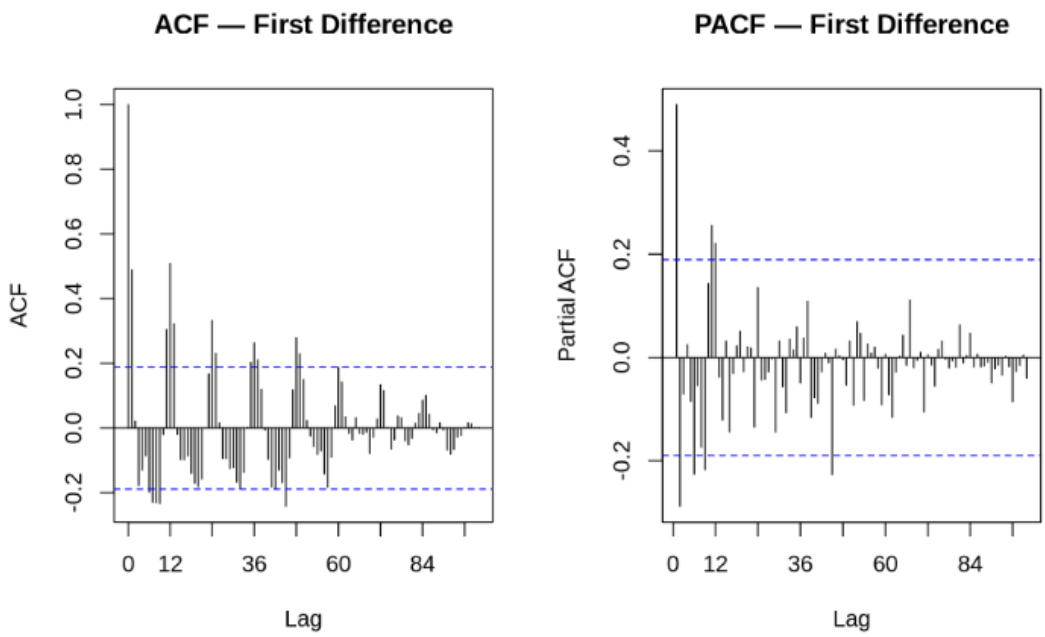

Figure 3: ACF and PACF of the first-differenced Box-Cox series. Seasonal spikes at multiples of lag 12 motivate a further seasonal difference.

The ACF and PACF of the doubly differenced series (Figure 4) are used to identify ARMA orders. The non-seasonal ACF cuts off after lag 1 while the PACF tails off, suggesting MA(1) ($q = 1$). An AR interpretation with $p \in \{1, 2\}$ was also retained as an alternative, since the PACF shows a clear early cut-off. At the seasonal frequency, the ACF cuts off after lag 12 while the PACF tails off, indicating a seasonal MA(1) term ($Q = 1$).

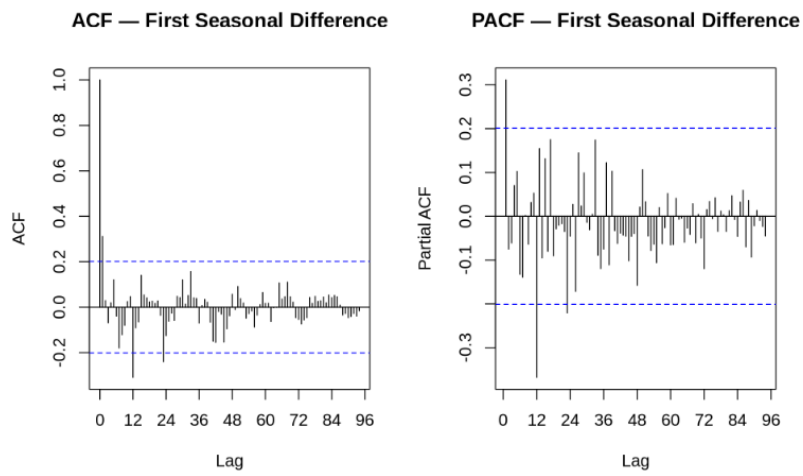

Figure 4: ACF and PACF of the doubly differenced series. Non-seasonal ACF cuts off at lag 1 ($q = 1$); seasonal ACF cuts off at lag 12 ($Q = 1$).

3. Fitting and Diagnostics

3.1. Candidate Models

Five candidate SARIMA models were estimated on the Box-Cox-transformed training series. Four were manually specified from the ACF/PACF reading; one was selected by `auto.arima` (full grid search, no approximation):

Label	Specification
`auto.arima`	$\text{ARIMA}(1,1,0)(0,1,1)_{12}$
M2	$\text{ARIMA}(2,1,0)(0,1,1)_{12}$
M3	$\text{ARIMA}(0,1,1)(0,1,1)_{12}$
M4	$\text{ARIMA}(0,1,1)(0,1,0)_{12}$

3.2. Model Selection

Table 2 reports AIC and BIC for all five models.

Table 2: Information criteria for all candidate models (lower is better).

Model	AIC	BIC
`auto.arima`: ARIMA$(1,1,0)(0,1,1)_{12}$	400.42	408.08
M2: ARIMA$(2,1,0)(0,1,1)_{12}$	400.54	410.75
M3: ARIMA$(0,1,1)(0,1,1)_{12}$	**398.84**	**406.50**
M4: ARIMA$(0,1,1)(0,1,0)_{12}$	428.21	433.32

M3 achieves the lowest AIC and BIC. M4 is dominated and dropped. The two finalists carried forward for residual diagnostics are M3 and `auto.arima`.

3.3. Residual Diagnostics

Table 3 summarises the three white-noise tests applied to the residuals of each finalist. A p-value > 0.05 is the desired outcome (fail to reject the null of white noise / normality / independence).

Table 3: Residual diagnostic p-values for the two finalist models.

Test	`auto.arima`	M3
Ljung-Box ($\ell = 20$)	0.456	0.690
Shapiro-Wilk	0.226	0.176
Runs Test	0.435	0.435

Both models pass all three tests, confirming residuals consistent with Gaussian white noise. ACF/PACF and Q-Q plots are in Figures 5–6.

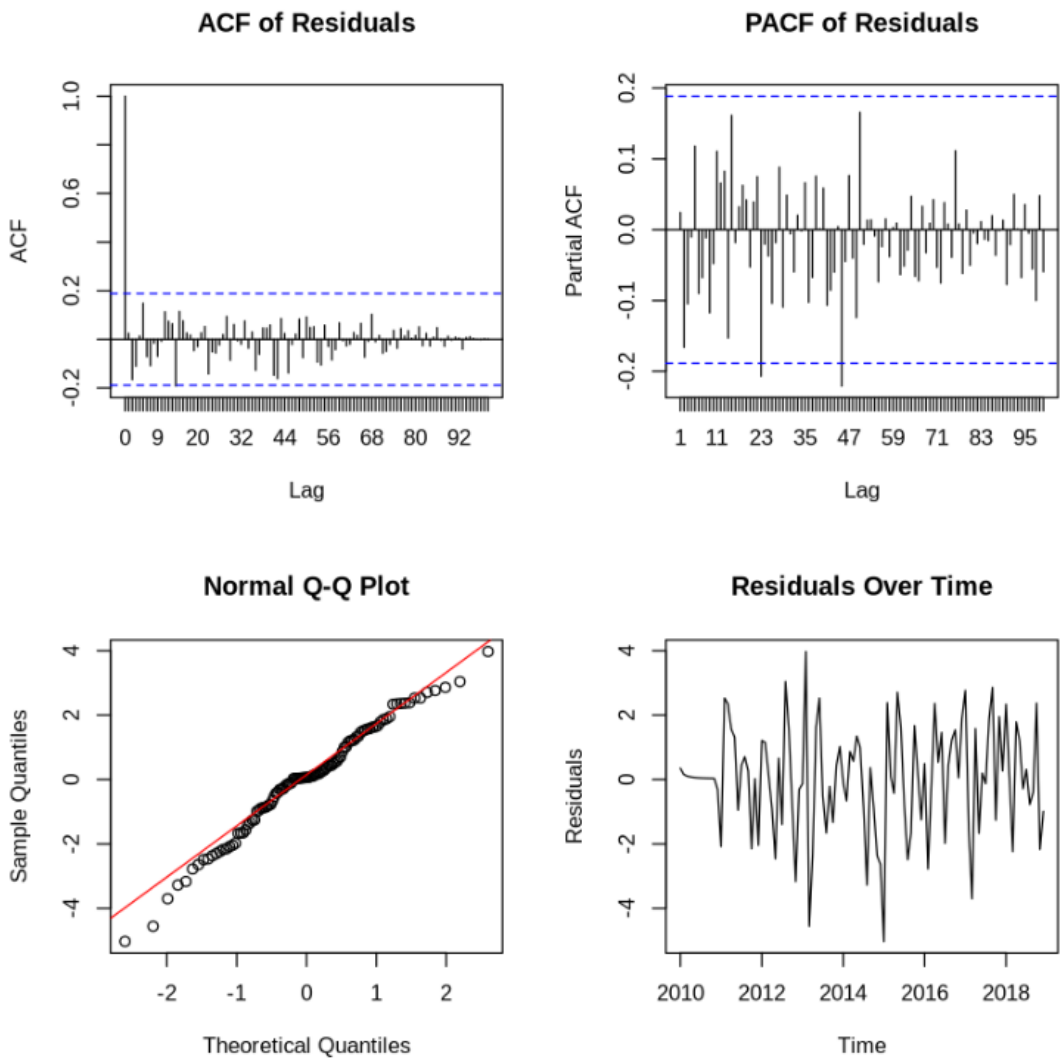

Figure 5: Residual diagnostics — `auto.arima`. No significant autocorrelation; Q-Q plot close to the normal line.

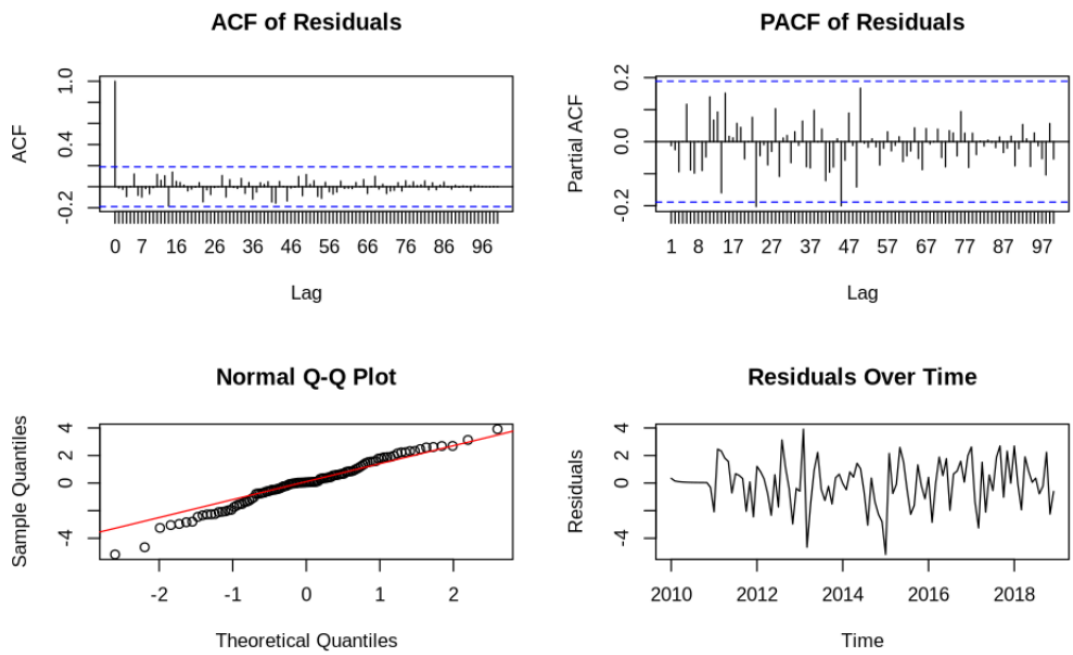

Figure 6: Residual diagnostics — M3. All lags within confidence bands; Q-Q plot confirms approximate normality.

3.4. Final Model and Possible Deficiencies

M3 — ARIMA$(0,1,1)(0,1,1)_{12}$ is selected as the final model. Its estimated equation is:

$$(1-B)(1-B^{12})\,y_t^{(\lambda)} = (1+0.4028\,B)(1-0.8921\,B^{12})\,\varepsilon_t, \quad \hat{\sigma}^2 = 3.042.$$

This is the well-known *airline model* (Box & Jenkins, 1970). The non-seasonal MA(1) coefficient ($\hat{\theta}_1 = 0.40$) captures month-to-month momentum in CPI changes: a price shock this month is partially passed through to the next month's index, reflecting real-world supply-chain lags and overlapping wage contracts. The seasonal MA(1) coefficient ($\hat{\Theta}_1 = -0.89$) encodes strong mean-reversion in the annual inflation cycle: an above-average price reading in a given calendar month tends to be followed by a below-average reading the following year, consistent with the stable inflation environment of 2010–2018.

Possible deficiencies. Theil's $U > 1$ on the test set shows the model is outperformed by a

seasonal naive benchmark at the full 12-step horizon — a known limitation of linear ARIMA models over long horizons. The model also assumes constant parameters; any structural shift in US inflation dynamics would require re-estimation. Finally, treating the Box-Cox parameter as fixed slightly understates forecast interval coverage.

4. Forecasting

Both finalist models — `auto.arima` and M3 — were used to generate 12-step-ahead point forecasts and 80%/95% prediction intervals for the CPI-U over January–December 2019, back-transformed to the original index scale via the inverse Box-Cox transformation.

Figures 7 and 8 overlay the forecasts and prediction bands against the actual 2019 CPI values. Both models trace the upward trend convincingly through 2019, and all 12 actual observations fall within the respective 95% intervals.

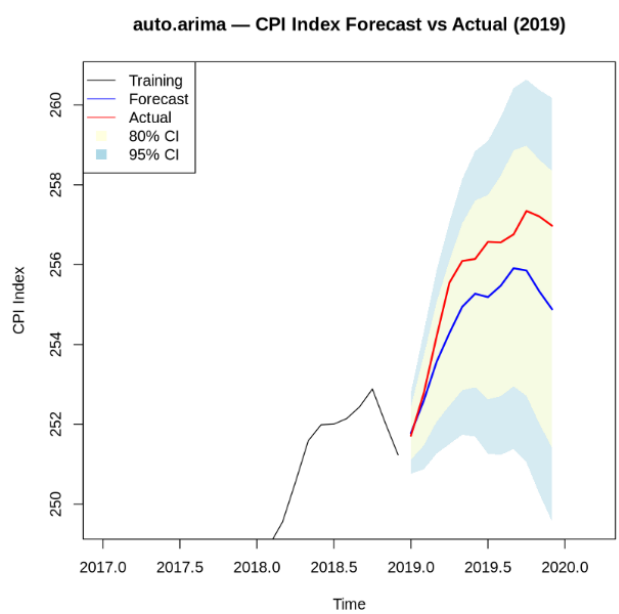

Figure 7: `auto.arima` ARIMA$(1,1,0)(0,1,1)_{12}$: forecast (blue) vs. actual 2019 CPI (red). All actual values lie within the 95% band.

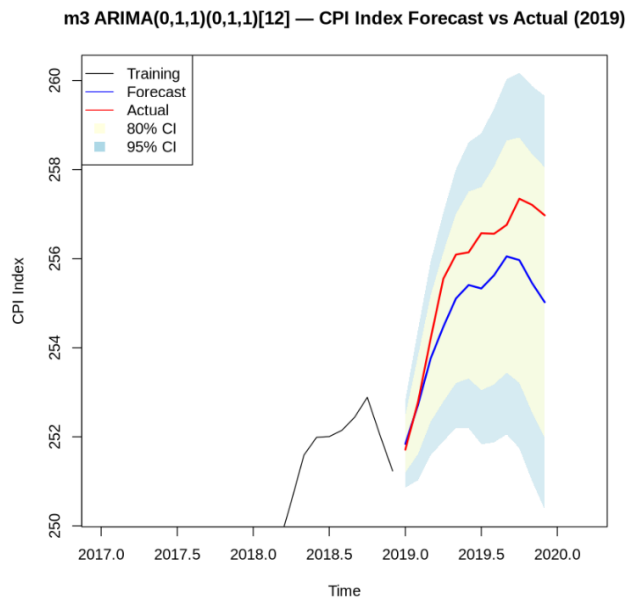

Figure 8: M3 ARIMA$(0,1,1)(0,1,1)_{12}$: forecast (blue) vs. actual 2019 CPI (red). M3 tracks the actual values more closely.

Table 4 reports out-of-sample accuracy on the 2019 test set.

Table 4: Out-of-sample forecast accuracy on the 2019 test set (12 obs.). Best values in bold.

Model	RMSE	MAE	MAPE (%)	MASE	Theil's U
`auto.arima`	1.226	1.079	0.421	0.074	1.726
M3	**1.102**	**0.950**	**0.370**	**0.066**	**1.551**

M3 outperforms `auto.arima` on every metric. A MAPE of 0.37% means that the model's predicted monthly CPI level differs from the actual value by less than four-tenths of one index point on average over the full forecast year — a strong result for a model estimated entirely on 2010–2018 data. The MASE below 1 confirms M3 beats the seasonal naive benchmark at the one-step level.

5. Discussion

This project set out to model and forecast the US CPI-U using standard SARIMA methodology. The central finding is that ARIMA$(0,1,1)(0,1,1)_{12}$ — the airline model — provides an excellent fit to monthly CPI data from a period of low, stable US inflation.

The analysis showed that the CPI trend cannot be removed deterministically: both a linear trend and a linear-plus-sinusoidal model left residuals that were non-stationary, non-normal, and serially dependent. This makes economic sense — CPI inflation is a continuous stochastic process, not a fixed deterministic path. Regular first differencing ($d = 1$) and seasonal differencing ($D = 1$) were both necessary to achieve stationarity, capturing the dual nature of CPI dynamics: a slow-moving drift in the price level and a recurring annual cycle in price changes.

The selected model M3 passed all residual diagnostic tests and achieved a MAPE of 0.37% on the unseen 2019 test year, outperforming the model automatically selected by `auto.arima` on every reported metric. Economically, the MA(1) coefficient reflects real-world price stickiness and supply-chain lags, while the seasonal MA(1) term captures the annual rhythm in price-setting behaviour — for example, seasonal energy costs, clothing cycles, and academic-year pricing.

Main challenges encountered. The most consequential judgment call was the choice of training window. The full CPI-U series runs from 1913, but using it in its entirety would have introduced multiple structural breaks (post-war inflation spikes, 1970s stagflation, the Volcker disinflation) that a single ARIMA model cannot accommodate. Restricting the sample to 2010–2018 sacrifices historical depth but produces a stable, homogeneous estimation environment. A natural extension for future work would be to apply regime-switching or time-varying parameter models to the full century-long series, or to incorporate leading economic indicators — such as producer prices or import costs — as explanatory variables.

References

[1] DataHub (2026). *US Consumer Price Index and Inflation (CPI)*. Available at: `https://datahub.io/core/cpi-us`